Begena Practice Coach

Product Requirements Document (PRD). Milestone 1.

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Milestone	1
Platforms	iOS, Android
Build approach	React Native UI with native audio and analysis module
Primary outcome	Daily home practice with trusted feedback
Session length target	8 to 12 minutes

1. Overview

Begena Practice Coach helps beginners practice at home with structure and instant feedback.

Milestone 1 focuses on finger drills that use two or three strings. The app guides a short daily routine and scores the attempt.

2. Goals

Build a daily habit for beginners.

Give feedback that feels fair.

Improve timing stability and reduce wrong-string hits in simple drills.

3. Target users

Beginners at Begena training institutes.

Remote beginners who practice alone and need guidance.

4. Problem statement

Many students stop because home practice breaks.

Finger drills feel boring. Students skip them or rush them.

Remote learners practice offline with no trusted progress check.

5. Product promise

One daily session. Clear steps. Fast setup.

Scores for timing and correct strings in two or three string drills.

Actionable feedback and a focused retry loop.

6. Scope for Milestone 1

6.1 Daily session flow

The session runs in three short sections, in order.

Section	Goal	Time target
Quick tune	Tune only strings used today. Capture string signatures for validation.	60 to 90 seconds
Finger drill	Repeat a two or three string pattern with a tempo ramp. Record and score.	5 to 8 minutes
Ear check	Identify which string played. Limited to today’s strings.	60 to 90 seconds

6.2 Drill constraints

Milestone 1 includes 10 to 20 beginner drills.

Each drill uses two or three strings.

Each drill uses a timed tempo ramp, slow to faster, like in physical training.

6.3 Non-goals

Full mezmur teaching.

Automatic validation for complex multi-string patterns.

Teacher review workflows.

Social features, discovery feeds, or large content catalogs.

7. Functional requirements

7.1 Quick tune

User selects target ቅኝት preset for their level.

App guides tuning for only strings used in today’s drill.

App plays a reference tone per string and shows a tuner indicator with tolerance.

When a string stays in range, the app saves a short open-string signature for that string.

7.2 Finger drill player

Shows the repeating string-number sequence.

Metronome, count-in, tempo ramp, and loop controls.

Records one scored attempt per session.

Offers focused retry for weak segment or weak steps.

7.3 Validation and scoring

Timing validation uses pluck onset timestamps compared to the beat grid.

String validation uses signatures captured during tuning and matches each pluck to the closest string signature.

Confidence gating controls string feedback. Low confidence hides string claims and prompts user to improve setup.

Consistency metrics include timing variance per segment and longest clean repetition streak.

Score type	What it measures	User output
Timing	Early or late vs beat grid, per strike and per segment	Segment score plus specific timestamps to re
Strike count	Missed strikes and extra strikes	Counts plus where it occurred
String accuracy	Correct string hit for expected step, only if confidence is high	Wrong-string steps highlighted
Consistency	Timing variance and the clean repetition streak	Stability summary and next action

7.4 Ear check

App plays one of today’s tuned strings.

User selects which string number they heard.

Prompts focus only on today’s two or three strings.

7.5 Progress tracking

Daily streak. Minutes practiced. Best scores per drill.

Unlock rules based on timing and string accuracy where confidence is high.

8. Technical approach

React Native handles UI and session logic.

Native module handles metronome timing, audio capture, and analysis.

Metronome audio and recording share the same timing clock to avoid scoring drift.

Analysis runs on device. Milestone 1 stores scores and metadata locally.

Area	Approach
Onset detection	Detect pluck timestamps for timing and strike count
String matching	Use tuning-time signatures, match by spectral features
Features	Short-window spectral features such as MFCC and related summary stats
Classifier	Nearest-match against saved per-string signatures
Confidence	Thresholds drive when string feedback appears

9. Quality bars

Quick tune step completes in under 90 seconds for most users.

A 60 second drill returns results in under 2 seconds after recording ends.

String feedback only appears when confidence is high.

Practice works offline. Scores save locally.

10. Metrics

Day 1 to day 7 retention.

Sessions per week per user.

Average minutes per session.

Timing score improvement over first 7 sessions.

Wrong-string rate trend on two string drills when confidence is high.

Rate of low-confidence sessions and top causes.

11. Risks and mitigations

Risk: String detection feels wrong. Mitigation: constrain drills, tuning-time signatures, confidence gating, and timing-first feedback.

Risk: Tuning feels slow. Mitigation: tune only today’s strings and keep it guided.

Risk: Device audio latency breaks scoring. Mitigation: metronome and recording share one native timing clock.

12. Release plan

Internal test with one teacher and 5 students.

Pilot with 20 to 30 beginners for 2 weeks.

Ship Milestone 1 when users complete at least 5 sessions per week on average and feedback feels fair in most sessions.